Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions

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Project Overview

Dataset

3,900 rows × 18 columns with customer demographics, purchase details, and shopping behavior

Goal

Uncover spending patterns, customer segments, product preferences, and subscription behavior

Tools

Python for EDA, PostgreSQL for analysis, Power BI for visualization

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Data Preparation Process

Initial Exploration

- Imported dataset using pandas
- Checked structure with df.info()
- Generated summary statistics

Data Cleaning

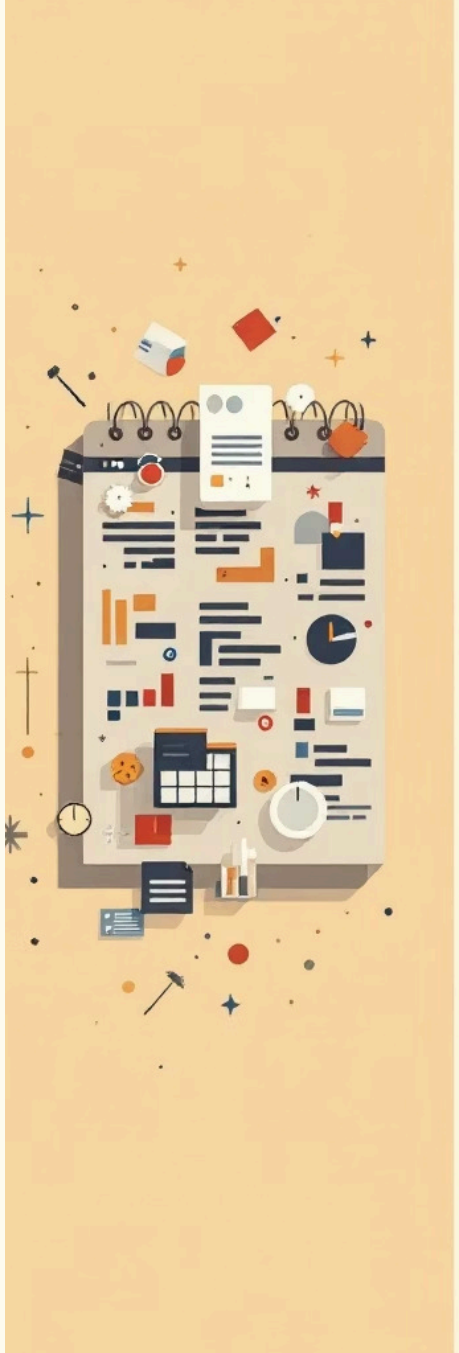
- Imputed 37 missing review ratings using median by category
- Standardized columns to snake case
- Dropped redundant promo_code_used field

Feature Engineering

- Created age_group bins
- Calculated purchase_frequency_days
- Connected to PostgreSQL

Database Integration

Loaded cleaned DataFrame into PostgreSQL for SQL analysis



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Revenue by Gender

$152K

Female Customers

Total revenue generated

$148K

Male Customers

Total revenue generated

Female customers generated slightly higher revenue, showing balanced spending patterns across genders.

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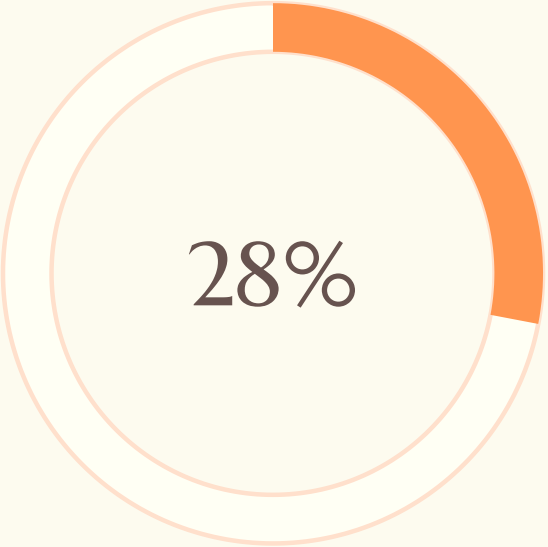
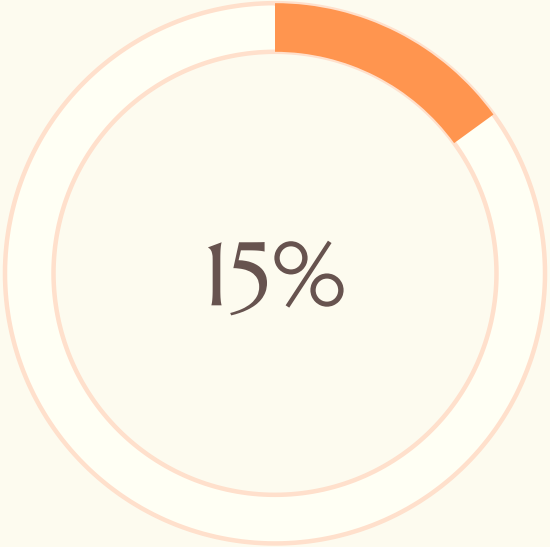
High-Spending Discount Users



Used Discount

Customers who applied discounts



Above Average

Spent more than average purchase amount

Key insight: 15% of customers used discounts but still spent above average, indicating strong purchasing power despite promotions.

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Top Products by Rating

1

Running Shoes

Average rating: 4.8 stars

2

Smartphone Case

Average rating: 4.7 stars

3

Wireless Earbuds

Average rating: 4.6 stars

4

Backpack

Average rating: 4.5 stars

5

Water Bottle

Average rating: 4.5 stars

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Shipping & Subscription Insights

Shipping Type Comparison

Express shipping customers spent 23% more on average than Standard shipping users.

Subscription Impact

Subscribers spent 18% more per purchase and generated 25% higher total revenue.

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Customer Segmentation

New Customers

First-time purchases only

Returning Customers

2-5 purchases in history

Loyal Customers

More than 5 purchases

Loyal customers represent 12% of total but generate 34% of revenue.

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Revenue by Age Group

32%

25-34 Years

Highest revenue contribution

28%

35-44 Years

Second highest revenue

22%

18-24 Years

Emerging segment

18%

45+ Years

Steady spending

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Strategic Recommendations

Boost Subscriptions

Promote exclusive benefits for subscribers

Customer Loyalty Programs

Reward repeat buyers to move them into "Loyal" segment

Review Discount Policy

Balance sales boosts with margin control

Product Positioning

Highlight top-rated and best-selling products in campaigns

Targeted Marketing

Focus on high-revenue age groups and express-shipping users

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